

Online Appendix

B Extensions

B.1 Subjective beliefs about private information

We begin with the case when players entertain subjective beliefs about their private information but have rational beliefs about the informativeness of the public signal. From equation (11), a player's anticipatory utility is always increasing in her subjective beliefs about the private signal. This implies that there always exists a symmetric equilibrium, as characterized by the following result.

Proposition 7. *Suppose all players have rational beliefs about the public signal i.e., $\delta_{\eta,i} = 1$. There exists a symmetric equilibrium in which all players are over-confident about their private signals and choose $\delta_{e,i}$, where:*

$$\delta_{e,i} = \begin{cases} \frac{2\psi+1}{2\psi-\frac{\tau_e}{\tau+\tau_\eta}} & \text{if } \psi > \frac{1+\bar{\delta}\frac{\tau_e}{\tau+\tau_\eta}}{2(\bar{\delta}-1)} \\ \bar{\delta} & \text{otherwise.} \end{cases} \quad (64)$$

Moreover, $\delta_{e,i}$ is (weakly) increasing in τ_e , and (weakly) decreasing in τ , τ_η , and ψ .

Consistent with the intuition from Section 4.2, players always want to distort their perception of private signal precision upwards. However, the experienced utility component of the objective now adds a “utility” cost to distorting beliefs. Intuitively, the more player i distorts her beliefs, the larger the expected loss in experienced utility (under the objective measure) from using subjective beliefs when choosing actions. Intuitively, this overestimation is larger when it is less costly to deviate from rational expectations (e.g., when prior information or public information is imprecise, i.e., low τ and low τ_η , or the cost of distorting beliefs is small, i.e., low ψ) and when the available private information is more precise (high τ_e). When the overall costs are sufficiently low (e.g., if ψ is sufficiently low), the player chooses the maximal distortion feasible i.e., $\delta_{e,i} = \bar{\delta}$.

Next, we allow players to entertain subjective beliefs about both their private and public signals, and assume that they maximize a weighted average of anticipatory and experienced utility. While we are not able to offer a complete characterization of equilibria for the entire parameter range, as in the earlier subsections, we establish sufficient conditions for the existence of symmetric and mixed equilibria of each type in the following results.

Proposition 8. *Suppose players choose beliefs about both private and public information and let*

$$B_{opt} = \max \left\{ \min \left\{ \frac{B_{RE}(2\psi+1) - \psi\bar{B}}{2\psi}, \bar{B} \right\}, \bar{B} \right\} \quad \Gamma = 1 + \psi \left(2 - \frac{\bar{B} + B_{opt}}{B_{RE}} \right) \quad (65)$$

$$\underline{B} = \frac{\underline{\delta}\tau_\eta}{\tau + \bar{\delta}\tau_e + \underline{\delta}\tau_\eta}, \quad \bar{B} = \frac{\bar{\delta}\tau_\eta}{\tau + \bar{\delta}\tau_e + \bar{\delta}\tau_\eta}, \quad B_{RE} = \frac{\tau_\eta}{\tau + \tau_e + \tau_\eta}. \quad (66)$$

Assume $\psi < \frac{\tau_e}{2(\tau + \bar{\delta}\tau_e)}$. Then, the following hold.

- (i) If $\chi > \frac{B_{opt}}{\bar{B}}\Gamma$, then there exists a unique equilibrium in which all players choose $\delta_{e,i} = \bar{\delta}$ and $\delta_{\eta,i} = \underline{\delta}$.

(ii) If $\chi < \frac{B}{B_{opt}}\Gamma$, then there exists a unique equilibrium in which all players choose $\delta_{e,i} = \bar{\delta}$ and

$$\delta_{\eta,i} = \min \left\{ \bar{\delta}, \frac{2\psi + 1 - \chi}{2\psi - \frac{(1-\chi)\tau_\eta}{\delta\tau_e + \tau}} \right\}. \quad (67)$$

(iii) For $\chi \in \left[\frac{B}{B_{opt}}\Gamma, \frac{B_{opt}}{B}\Gamma \right]$, then there exists a unique equilibrium in which a fraction λ of players choose $\delta_{e,i} = \bar{\delta}$ and $\delta_{\eta,i} = \frac{B_{opt}}{1-B_{opt}} \frac{(\bar{\delta}\tau_e + \tau)}{\tau_\eta}$ while the remaining players choose $\delta_{e,i} = \underline{\delta}$ and $\delta_{\eta,i} = \underline{\delta}$.

The above result shares features of the equilibrium characterization in our main analysis. Specifically, the condition on the relative importance of experienced utility i.e., $\psi < \frac{\tau_e}{2(\tau + \delta\tau_e)}$, ensures that in any equilibrium, players choose to maximally distort their beliefs about their private information i.e., set $\delta_{e,i} = \bar{\delta}$.³⁹ Given this, when χ is sufficiently small (part (i)), we show there exists a symmetric equilibrium in which all players exhibit over-confidence about their private information and over-react to public information. Similarly, when χ is sufficiently large (part (ii)), all players exhibit over-confidence in their private information, but under-react to public information. And for intermediate χ , there exists a unique mixed equilibrium in which some players over-react to the public signals, while others under-react to it.

B.1.1 Proof of Proposition 7

The objective of player i is

$$\max_{\delta_{e,i} \in (\underline{\delta}, \bar{\delta})} AU_i + \psi EU_i$$

The lower constraint is never binding (when players only choose $\delta_{e,i}$) and the objective can be rewritten as

$$\max_{\delta_{e,i}} AU_i + \psi EU_i \quad \text{subject to } \delta_{e,i} < \bar{\delta}$$

Let ω_e denote the Lagrange multiplier for the above inequality. We define the Lagrangian

$$L = -\frac{1}{\tau + \delta_{e,i}\tau_e\tau_\eta} + \chi \frac{B^2}{\tau_\eta} - \psi \frac{\tau + \delta_{e,i}^2\tau_e + \tau_\eta}{(\tau + \delta_{e,i}\tau_e + \tau_\eta)^2} - \omega_e(\delta_{e,i} - \bar{\delta}).$$

The FOC with respect to $\delta_{e,i}$ for any equilibrium is

$$\frac{\tau_e}{(\tau_\eta + \delta_{e,i}\tau_e + \tau)^2} \left(1 - 2\psi\delta_{e,i} + \frac{2\psi(\tau + \delta_{e,i}^2\tau_e + \tau_\eta)}{\tau + \delta_{e,i}\tau_e + \tau_\eta} \right) - \omega_e = 0. \quad (68)$$

Following steps similar to those in the proof of Proposition 2, it can be shown that the objective function has either a single interior maximum or is always increasing. Hence, the

³⁹While a small enough ψ is needed for the proof, numerical simulations show that the qualitative results hold more generally. Thus, we view the requirement of a small ψ in Proposition 8 as technical and not restrictive for the economic mechanism.

solution is

$$\delta_{e,i} \equiv \min \left(\frac{2\psi + 1}{2\psi - \frac{\tau_e}{\tau + \tau_\eta}}, \bar{\delta} \right). \quad (69)$$

B.1.2 Proof of Proposition 8

The objective of player i is

$$\max_{\delta_{\eta,i}, \delta_{e,i} \in [\underline{\delta}, \bar{\delta}]} -\frac{1}{\tau + \delta_{e,i}\tau_e + \delta_{\eta,i}\tau_\eta} + \chi \frac{B^2}{\delta_{\eta,i}\tau_\eta} - \psi \frac{\tau + \delta_{e,i}^2\tau_e + \delta_{\eta,i}^2\tau_\eta}{(\tau + \delta_{e,i}\tau_e + \delta_{\eta,i}\tau_\eta)^2}.$$

Let's first focus on the choice of $\delta_{e,i}$. Let $\omega_{e,1}$ and $\omega_{e,2}$ denote the Lagrange multipliers for the inequalities $\underline{\delta} < \delta_{e,i}$ and $\delta_{e,i} < \bar{\delta}$ respectively. We define the Lagrangian

$$L = -\frac{1}{\tau + \delta_{e,i}\tau_e + \delta_{\eta,i}\tau_\eta} + \chi \frac{B^2}{\delta_{\eta,i}\tau_\eta} - \psi \frac{\tau + \delta_{e,i}^2\tau_e + \delta_{\eta,i}^2\tau_\eta}{(\tau + \delta_{e,i}\tau_e + \delta_{\eta,i}\tau_\eta)^2} + \omega_{e,1}\delta_{e,i} - \omega_{e,2}\delta_{e,i}.$$

The FOC with respect to $\delta_{e,i}$ for any equilibrium is

$$\frac{\tau_e}{(\delta_{\eta,i}\tau_\eta + \delta_{e,i}\tau_e + \tau)^2} \left(1 - 2\psi\delta_{e,i} + 2\psi \frac{(\tau + \delta_{e,i}^2\tau_e + \delta_{\eta,i}^2\tau_\eta)}{\tau + \delta_{e,i}\tau_e + \delta_{\eta,i}\tau_\eta} \right) + \omega_{e,1} - \omega_{e,2} = 0.$$

If $\psi < \frac{\tau_e}{2(\tau + \bar{\delta}\tau_e)}$, the constraint $\delta_{e,i} \leq \bar{\delta}$ is binding (irrespective of the choice of $\delta_{\eta,i}$) and all players choose $\delta_{e,i} = \bar{\delta}$. The rest of the proof follows from proposition 2 with τ_e replaced with $\bar{\delta}\tau_e$.

B.2 Subjective beliefs about others' signals

In this section, we modify our benchmark model to allow players to hold subjective beliefs about the private signals observed by other players. As in the benchmark model, each individual observes

$$s_i = \theta + \varepsilon_i \quad \varepsilon_i \sim N(0, 1/\tau_e), \quad (70)$$

but player i believes that

$$\varepsilon_j =_i \sqrt{1 - p_i^2\eta} + p_i\varepsilon_j \quad \varepsilon_j \sim N\left(0, \frac{1}{\delta_i\tau_e}\right), \quad (71)$$

where p_i and δ_i are chosen by player i to maximize her anticipatory utility, net of costs. To highlight the role of beliefs about others, we shut down the public signal, s , and we do not allow players to hold subjective beliefs about the precision of their own signal. All other features of the benchmark model are unchanged.

Suppose player i conjectures that $K = \kappa_0 + \alpha \left(\theta + \sqrt{1 - p_i^2\eta} \right)$, where α is determined in equilibrium. Then,

$$k_i = \kappa_0 + (r\alpha + (1 - r)\kappa_1) \mathbb{E}_i[\theta|s_i] \quad (72)$$

This implies that, in equilibrium, player i believes that the average action (across all players) is given by

$$K = \int_i k_i di = \kappa_0 + (r\alpha + (1 - r)\kappa_1) A \left(\theta + \sqrt{1 - p_i^2\eta} \right), \quad (73)$$

Matching terms, we show that α is unchanged from our benchmark model. Players' subjective belief about the private signals of others is given by the following proposition.

Proposition 9. *Suppose players choose subjective beliefs about the private signals of others to maximize a weighted average of anticipatory and experienced utility (as in (6)). Then,*

- (i) *If $u_{\sigma\sigma} \geq u_{KK}$, all players choose $p_i = 1$. Moreover, players choose $\delta_i = \underline{\delta}$ iff $u_{\sigma\sigma} > 0$, and choose $\delta_i = \bar{\delta}$ otherwise.*
- (ii) *If $u_{\sigma\sigma} < u_{KK}$, all players choose $p_i = 0$. Moreover, players choose $\delta_i = \underline{\delta}$ iff $u_{KK} > 0$, and choose $\delta_i = \bar{\delta}$ otherwise.*

Proof. We can rewrite anticipatory utility as

$$AU_i(p_i, \delta_i) = \mathbb{E}[u_0] + \mathcal{A}\text{var}(\theta) + \frac{1}{2}u_{\sigma\sigma}\mathbb{E}_i[\sigma_k^2] + \frac{1}{2}(u_{kk}\mathbb{E}_i[(k - \mathcal{K})^2] + (u_{KK} - r^2u_{kk})\mathbb{E}_i[(K - \kappa)^2]), \quad (74)$$

where the key distinction from our benchmark is that players' subjective beliefs about other's signals impact their expectation of the dispersion in their actions, σ_k^2 . Given player i 's subjective beliefs,

$$\mathbb{E}_i[\sigma^2] = \int (k_i - K)^2 di = \frac{p_i^2 \alpha^2}{\delta_i \tau_e} \quad (75)$$

$$\mathbb{E}_i[(k - \mathcal{K})^2] = \frac{(r\alpha + (1-r)\kappa_1)^2}{\tau} + \frac{r^2 \alpha^2 (1 - p_i^2)}{\delta_i \tau_e} \quad (76)$$

$$\mathbb{E}_i[(K - \kappa)^2] = \frac{\left(\frac{\kappa_1(A-1)}{1-rA}\right)^2}{\tau} + \frac{\alpha^2 (1 - p_i^2)}{\delta_i \tau_e} \quad (77)$$

This implies that

$$\frac{\partial AU_i}{\partial \delta_i} = -u_{\sigma\sigma} \left(\frac{p_i^2 \alpha^2}{2\delta_i^2 \tau_e} \right) - u_{KK} \left(\frac{\alpha^2 (1 - p_i^2)}{2\delta_i^2 \tau_e} \right) \quad (78)$$

$$= -\frac{\alpha^2}{2\delta_i^2 \tau_e} (p_i^2 u_{\sigma\sigma} + (1 - p_i^2) u_{KK}) \quad (79)$$

$$\frac{\partial AU_i}{\partial p_i} = u_{\sigma\sigma} \left(\frac{p_i \alpha^2}{\delta_i \tau_e} \right) - u_{KK} \frac{\alpha^2 p_i}{\delta_i \tau_e} \quad (80)$$

$$= \frac{p_i \alpha^2}{\delta_i \tau_e} (u_{\sigma\sigma} - u_{KK}) \quad (81)$$

Finally, note that since player i 's action does not depend on p_i or δ_i , their experienced utility is also independent of p_i or δ_i . Hence, we have the result. $\square$

B.3 Subjective beliefs about means

In this section, we modify our benchmark model to allow players to hold subjective beliefs about the mean of both their own private signal as well as the public signal. Specifically, player i believes the errors in her signals are

$$\varepsilon_i \sim_i N\left(\mu_{e,i}, \frac{1}{\tau_e}\right), \quad \eta \sim_i N\left(\mu_{\eta,i}, \frac{1}{\tau_\eta}\right).$$

Given player i 's subjective beliefs, $\mu_{e,i}$ and $\mu_{\eta,i}$, and the realization of s_i and s , her optimal action, k_i is

$$k_i^*(\mu_{e,i}, \mu_{\eta,i}) = \mathbb{E}_i[rK + (1-r)\kappa(\theta) | s_i, s] = r\mathbb{E}_i[K | s_i, s] + (1-r)\mathbb{E}_i[\kappa(\theta) | s_i, s],$$

Given our assumptions on the joint distribution of fundamentals and signals, Bayesian updating implies that player i 's conditional beliefs about θ are given by

$$\mathbb{E}_i[\theta | s_i, s] = A(s_i - \mu_{e,i}) + B(s - \mu_{\eta,i}), \text{ and } \text{var}_i[\theta | s_i, s] = \frac{1 - A - B}{\tau},$$

where player i 's weights on the private and public signals are given by:

$$A \equiv \frac{\tau_e}{\tau + \tau_e + \tau_\eta}, \text{ and } B \equiv \frac{\tau_\eta}{\tau + \tau_e + \tau_\eta}.$$

With this, we establish the existence of a unique equilibrium, given players' beliefs.⁴⁰

Lemma 4. *Given a choice of subjective beliefs $\{\mu_{e,i}, \mu_{\eta,i}\}_i$ for all players, there always exists a unique, linear equilibrium in which player i 's optimal action is given by:*

$$k_i(\mu_{e,i}, \mu_{\eta,i}) = \kappa_0 + r\gamma_e\mu_e + r\gamma_\eta\mu_\eta + r\beta\mu_{\eta,i} + \alpha(s_i - \mu_{e,i}) + \beta(s - \mu_{\eta,i}),$$

and the aggregate action is given by:

$$K = \kappa_0 + \gamma_e\mu_e + \gamma_\eta\mu_\eta + \alpha\theta + \beta s,$$

$$\text{where } \alpha = \frac{A(1-r)}{1-rA}\kappa_1, \beta = \frac{B}{1-rA}\kappa_1, \mu_e = \int \mu_{e,i} di, \mu_\eta = \int \mu_{\eta,i} di, \gamma_e = -\frac{\kappa_1 A}{1-rA}, \gamma_\eta = -\frac{\kappa_1 B}{1-rA}$$

With the equilibrium established, we then establish how $\mu_{e,i}$ and $\mu_{\eta,i}$ affect both players' anticipatory utility as well as the cost borne (through the experienced utility penalty).

Proposition 10. *Given player i 's subjective beliefs, $\mu_{e,i}$ and $\mu_{\eta,i}$, anticipatory utility can be written*

$$AU_i(\mu_{e,i}, \mu_{\eta,i}) = \Gamma + u_K\beta(\mu_{\eta,i} - \mu_\eta) + \frac{1}{2}(u_{KK} - r^2u_{kk})(\gamma_e\mu_e + \beta(\mu_{\eta,i} - \mu_\eta))^2 \quad (82)$$

and the experienced utility penalty is

$$EU_i(\mu_{e,i}, \mu_{\eta,i}) = u_{kk}(\alpha\mu_{e,i} + \beta(1-r)\mu_{\eta,i})^2$$

Players' anticipatory utility doesn't depend on $\mu_{e,i}$: intuitively, shifting the mean of the private signal is "accounted for" when updating and so player i 's anticipatory utility is unaffected. However, since the true mean is zero, choosing $\mu_{e,i} \neq 0$ is objectively costly. As a result, all players choose $\mu_{e,i} = 0$.

Thus, we rewrite the objective of player i :

$$\max_{\mu_{\eta,i}} AU_i + \psi EU_i.$$

First, we characterize the equilibrium when aggregate errors volatility decreases anticipatory utility, i.e., when $\chi < 0$. In order to characterize the equilibrium, we assume that $\mu_{\eta,i} \in [-\bar{\mu}, \bar{\mu}]$. This is similar to the set of reasonable beliefs imposed in our baseline analysis.

Proposition 11. *If aggregate errors volatility decreases anticipatory utility (i.e., $\chi \leq 0$), then there exists a unique symmetric equilibrium in which*

$$\mu_e = 0 \quad \mu_{\eta,i} = \mu_\eta = \min\left(\max\left(\frac{u_K}{-2u_{kk}\psi\beta(1-r)^2}, -\bar{\mu}\right), \bar{\mu}\right) \text{ for all } i \in [0, 1]. \quad (83)$$

⁴⁰All the proofs for this section follow similar steps as in the main model and are omitted for brevity.

Next, we consider the case when aggregate errors volatility increases anticipatory utility.

Proposition 12. *If aggregate errors volatility increases anticipatory utility (i.e., $\chi > 0$), then there exists a $\bar{\chi} > 0$ such that:*

(i) *If $\chi > \bar{\chi}$, the unique equilibrium is mixed. In this equilibrium, a fraction λ of players optimally chooses $\mu_{e,i} = 0$ and $\mu_{\eta,i} = -\bar{\mu}$, while the remaining fraction $1 - \lambda$ optimally chooses $\mu_{e,i} = 0$ and $\mu_{\eta,i} = \bar{\mu}$.*

(ii) *If $\chi < \bar{\chi}$, then the unique equilibrium is symmetric and is given by (83).*

While players never choose to distort their beliefs about the mean of their private signals, players always choose to distort their beliefs about the mean of the public signal. Specifically, we find that there exists a symmetric equilibrium when either the effect of aggregate errors volatility on anticipatory utility is not too high. In this symmetric equilibrium, whether players choose to believe that the public signal is biased upward or downward (i.e., positive or negative μ_{η} , respectively) depends upon whether the aggregate action, u_K , imposes a positive or a negative externality. On the other hand, when non-fundamental volatility increases anticipatory utility sufficiently, there does not exist a symmetric equilibrium - instead, players endogenously choose to disagree about the mean of the public signal.

C Applications

In this section, we consider the implications of our analysis for a few applied models.

C.1 Efficient Competitive Economies

Consider an incomplete-market competitive economy in which agents' choices are strategic substitutes. There are two goods and a continuum of households (who act as consumers and producers). Let q_{1i} and q_{2i} denote the respective quantities purchased by consumer i (the consumer living in household i). The preferences of this consumer are given by

$$u_i = \theta q_{1i} - \frac{b q_{1i}^2}{2} + q_{2i}$$

where the random variable θ represents a shock in the relative demand for the two goods and the budget is

$$p q_{1i} + q_{2i} = e + \pi_i$$

where p is the price of good one, good two serves as numeraire, e is an exogenous endowment of good 2, and π_i are the profits of producer i . Profits in turn are given by $\pi_i = p k_i - \frac{k_i^2}{2}$ where k_i denotes the quantity of good 1 produced by household i .

Consumer i chooses the optimal bundle (q_{1i}, q_{2i}) so as to maximize utility subject to budget constraint, which gives $p = \theta - b q_{1i}$. Households are ex-ante identical which, together with market clearing, implies that $q_{1i} = K$ for all i and therefore $p = \theta - bK$ where $K = \int k_i di$. This example is thus nested in our model with the utility

$$U(k, K, \sigma_k, \theta) = (\theta - bK) k - k^2/2 + bK^2/2 + e$$

which implies that

$$u_{kk} = -1, u_{kK} = -b, u_{k\theta} = 1, u_{KK} = b$$

and $\kappa_0 = 0; \kappa_1 = \frac{1}{1+b}$ and

$$\chi \equiv \frac{u_{KK} - u_{kk}r^2}{-u_{kk}(1-r)^2} = \frac{b}{1+b} > 0.$$

Since $\chi \in (0, 1)$ increases in b , Proposition 2 shows that whether the equilibrium belief choice is symmetric or mixed depends upon the level of b . If b is too low, then the equilibrium is symmetric. For high enough b , the equilibrium is mixed. Proposition 3 implies that, in symmetric equilibrium (low b), the players overreact to public information. In a mixed equilibrium, players under-react to information if $\chi = \frac{b}{1+b} > \frac{\Gamma_{BB_{opt}}}{B_{RE}^2}$ i.e., if b is high. Moreover, the average weight on public information decreases in b .

Proposition 4 implies that, in a symmetric equilibrium (i.e., low b), dispersion of beliefs increases with b . In a mixed equilibrium (when b is high), the dispersion of beliefs is hump-shaped in b . Proposition 5 implies that, in a symmetric equilibrium, there is over-reaction at the individual level and there will be under-reaction at the consensus level if either the set of reasonable beliefs is small or if b is high. Moreover, both CG_i and CG_a are increasing in b . Proposition 7, implies that, when b is high, the dispersion of individual CG coefficients is hump-shaped in χ .

C.2 Investment complementarities

Consider a setting in which the terminal value of the firm is given by $V \equiv V(R, k)$ where R measures the return on investment, or productivity, of the project available and k represents the scale of investment in the project. For analytical tractability, let

$$V(R, k) = Rk - \frac{1}{2}k^2, \quad (84)$$

$$R = (1-a)\theta + aK, \quad (85)$$

where $\theta \in R$ represents the firm's exogenous productivity. This implies that firm i 's utility is

$$U_i(k_i, K, \theta) = (1-a)\theta k_i + aKk_i - \frac{1}{2}k_i^2$$

which falls into the class of general objective functions we analyzed. This implies that

$$u_{kk} = -1, u_{kK} = a, u_{k\theta} = 1-a.$$

which implies that $r = a$ and $\kappa_1 = 1$ and so

$$\chi = -\frac{(u_{KK} - r^2 u_{kk})}{u_{kk}(1-r)^2} = \frac{a^2}{(1-a)^2} > 0$$

In this economy, χ increases with a . Moreover, $\chi < 1 \iff a < \frac{1}{2}$. Proposition 2 shows that whether the equilibrium belief choice is symmetric or mixed depends upon the level of a . If a is too low or too high, then the equilibrium is symmetric. For medium a , the equilibrium is mixed. Proposition 3 implies that, in symmetric equilibrium, the players under-react to public information if $a > \frac{1}{2}$. In a mixed equilibrium, players under-react to information if $\chi = \frac{a}{1-a} > \frac{\Gamma_{BB_{opt}}}{B_{RE}^2}$ i.e., if a is high enough. Moreover, the average weight on public information decreases in a .

Proposition 4 implies that, in a symmetric equilibrium, the dispersion of beliefs increases with a . In a mixed equilibrium, the dispersion of beliefs is hump-shaped in a . Proposition 5 implies that, in a symmetric equilibrium, there is over-reaction at the individual level if

$a < \frac{1}{2}$ and there will be under-reaction at the consensus level if either the set of reasonable beliefs is small or if a is high. Moreover, both CG_i and CG_a are increasing in a . Proposition 7, implies that, in a mixed equilibrium, the dispersion of individual CG coefficients is hump-shaped in a .

C.3 Cournot versus Bertrand

We next study two IO applications with a large number of firms: a Cournot-like game, where firms compete in quantities and actions are strategic substitutes; and a Bertrand-like game, where firms compete in prices and actions are strategic complements.

First, consider a Cournot setting in which firms compete on quantity. For any firm, consumer demand is $p = a_0 + a_1\theta - a_2q - a_3Q$ (with $a_0, a_1, a_2, a_3 > 0$), where p denotes the price at which the firm sells each unit of its product, q is the quantity it produces, Q is the average quantity produced across all firms, and θ is a fundamental demand shock/shifter. Each firm's profits are given by $u = pq - C(q)$, where $C(q) = c_1q + c_2q^2$ is the cost function (with $c_1, c_2 > 0$). This model is nested in our general framework with $k \equiv q$, $K \equiv Q$, and

$$U(k, K, \sigma_k, \theta) = (a_0 - c_1 + a_1\theta - a_3K)k - (a_2 + c_2)k^2.$$

This implies that $U_{kk} = -2(a_2 + c_2)$, $U_{kK} = -a_3$, $U_{k\theta} = a_1$, $U_k = a_0 - c_1$ which in turn implies that $r = \frac{-a_3}{2(a_2 + c_2)}$, $\kappa_1 = -\frac{a_1}{-2(a_2 + c_2) - a_3}$ and

$$\chi = -\frac{(u_{KK} - r^2 u_{kk})}{u_{kk}(1 - r)^2} = \frac{\frac{a_3^2}{4(a_2 + c_2)^2}}{\left(1 + \frac{a_3}{2(a_2 + c_2)}\right)^2} > 0$$

which implies that $\chi \in (0, 1)$ and is increasing in $b \equiv \frac{a_3}{2(a_2 + c_2)}$. This, in turn implies that χ increases in a_3 but decreases in a_2 and c_2 .

Next, consider a Bertrand setting in which firms compete on price. Consumer demand for each firm is given by $q = b_0 + b_1\theta' - b_2p + b_3P$, where q denotes the quantity sold by the firm, p is the price the firm sets, P is the average price in the market, and θ' is an exogenous demand shifter ($b_0, b_1, b_2, b_3 > 0$); we naturally impose $b_3 < b_2$, so that an equal increase in p and P reduces q . Individual profits are $u = pq - C(q)$, where $C(q) = c_1q + c_2q^2$ (with $c_1, c_2 > 0$). This model is nested in our framework with $k \equiv p - c_1$, $K \equiv P - c_1$ (actions are now prices), and

$$U(k, K, \sigma_k, \theta) = \theta k(1 + 2c) - k^2(1 + c) + Kk(b + 2bc) - c\theta^2 - cb^2K^2 - 2bc\theta K$$

where $\theta = \frac{b_0}{b_2} + \frac{b_1}{b_2}\theta' - c_1(1 - b)$, $b \equiv \frac{b_3}{b_2}$ and $c \equiv c_2b_2$. This implies that $U_{kk} = -2(1 + c)$, $U_{kK} = b(1 + 2c)$, $U_{k\theta} = 1 + 2c$, $U_{KK} = -2cb^2$, $U_{K\theta} = -2bc$, $U_{\theta\theta} = -2c$ which in turn implies that $r = \frac{b(1 + 2c)}{2(1 + c)}$, $\kappa_1 = \frac{1 + 2c}{2(1 + c) - b(1 + 2c)}$ and

$$\chi = -\frac{(u_{KK} - r^2 u_{kk})}{u_{kk}(1 - r)^2} = \frac{b^2}{(2(1 + c) - b(1 + 2c))^2} > 0.$$

Since $b < 1$, one can show that $\chi \in (0, 1)$. Moreover, χ is increasing in $b = b_3/b_2$ and decreases in $c = c_2b_2$. This implies that χ increases in b_3 , decreases in b_2 and c_2 .

C.4 Information Spillovers

The model of [Angeletos, Lorenzoni, and Pavan \(2018\)](#) considers a novel channel through which the information in the real sector affects behavior in the financial sector. The real

sector is comprised of entrepreneurs making investment decisions, and the financial sector is comprised of investors who provide liquidity to the “real” economy. All players are risk-neutral and the discount rate is zero. Time is divided in three periods, $t \in \{1, 2, 3\}$. At $t = 1$, a new investment opportunity becomes available with productivity $\theta \sim N\left(0, \frac{1}{\tau_\theta}\right)$. This investment pays off at $t = 3$. There is a continuum of entrepreneurs who can choose how much to invest in new technology. Let k_i denote the investment of entrepreneur i , and let the cost of this investment be $\frac{k_i^2}{2}$. Entrepreneurs have access to information technology that generates both a private and a “public” signal that they utilize when making their investment decision.⁴¹ The joint distribution of fundamentals and signals follows the specification detailed in Section 3.

At $t = 2$, the “financial market” operates: some entrepreneurs transfer their capital to the traders. Each entrepreneur is hit by an idiosyncratic shock with probability $l \in [0, 1]$. Entrepreneurs hit by this shock do not value consumption at $t = 3$ and have no choice but to sell all their capital at $t = 2$, at a price p , to investors. For simplicity, the entrepreneurs not hit by the shock are precluded from trading. Entrepreneur preferences are given by $u_i = c_{i1} + c_{i2} + s_i c_{i3}$, where c_{it} denotes agent i ’s consumption in period t , while s_i is a random variable that takes value 0 if the agent is hit by a liquidity shock and value 1 otherwise. Each entrepreneur’s expected utility at the time of investment is given by

$$\mathbb{E}_i [u_i | s_i, s] = \mathbb{E}_i \left[(1-l) \theta k_i + l p k_i - \frac{1}{2} k_i^2 | s_i, s \right]. \quad (86)$$

The financial market is competitive and the market-clearing price is denoted by p . Investors do not have access to their own information technology but, given the assumptions above, update their beliefs about the productivity of the technology utilizing the information contained in the supply of capital to be liquidated. It can be shown that, given the distributional assumptions and the risk-neutrality of traders, that $p = \mathbb{E}[\theta | K] = \alpha_1 K$, where α_1 is pinned down in equilibrium. This implies that expected utility of entrepreneur is given by

$$\mathbb{E} [u_i (k_i, K, \theta) | s_i, s] = \mathbb{E}_i \left[(1-l) \theta k_i + l k_i \alpha_1 K - \frac{1}{2} k_i^2 | s_i, s \right]. \quad (87)$$

and therefore entrepreneurs optimal action is given by $k_i = \mathbb{E}_i [(1-l) \theta + l \alpha_1 K]$. Note that this objective falls into the class of general objective functions we studied. Aggregating across all entrepreneurs, this implies that aggregate investment can be written as

$$K = \frac{1-l}{1-lA} \kappa_1 A \theta + \frac{B}{1-lA} \kappa_1 s \quad (88)$$

$$= \frac{\kappa_1 ((1-l) A + B)}{1-lA} \left(\theta + \frac{B}{(1-l) A + B} \eta \right). \quad (89)$$

It is straightforward to see that the aggregate level of capital reveals a signal of the form $\xi = \theta + \frac{B}{((1-l)A+B)} \eta$ to investors which, given the linear-normal structure, verifies the

⁴¹It is public in the sense that all entrepreneurs observe the same signal; however, as we discuss below, it is assumed that investors do not observe the signal and can only learn about θ through the aggregate investment level.

conjectured functional form for the price of capital. This implies that α_1 solves the equation

$$\frac{\alpha_1 (1 - l)}{1 - l\alpha_1} = \frac{((1 - l\alpha_1) A + B) \tau_\eta}{((1 - l\alpha_1) A + B)^2 \tau_\eta + B^2 \tau_\theta} (1 - l\alpha_1 A)$$

On the other hand, however, note that

$$\chi \equiv \frac{u_{KK} - u_{kk} r^2}{-u_{kk} (1 - r)^2} = \frac{l^2 \alpha_1^2}{(1 - l\alpha_1)^2} > 0.$$

Note that χ increases with l . Moreover, $\chi < 1 \iff l\alpha_1 < \frac{1}{2}$. Proposition 2 shows that whether the equilibrium belief choice is symmetric or mixed depends upon the level of $l\alpha_1$. If $l\alpha_1$ is too low or too high, then the equilibrium is symmetric. For medium $l\alpha_1$, the equilibrium is mixed. Proposition 3 implies that, in symmetric equilibrium, the players under-react to public information if $l\alpha_1 > \frac{1}{2}$. In a mixed equilibrium, players under-react to information if $\chi = \frac{l\alpha_1}{1-l\alpha_1} > \frac{\Gamma_{BB_{opt}}}{B_{RE}^2}$ i.e., if l is high enough. Moreover, the average weight on public information decreases in l .

Proposition 4 implies that, in a symmetric equilibrium, the dispersion of beliefs increases with l . In a mixed equilibrium, the dispersion of beliefs is hump-shaped in l . Proposition 5 implies that, in a symmetric equilibrium, there is over-reaction at the individual level iff $l\alpha_1 < \frac{1}{2}$ and there will be under-reaction at consensus level if either the set of reasonable beliefs is small or if l is high. Moreover, both CG_i and CG_a are increasing in l . Proposition 6, implies that, in a mixed equilibrium, the dispersion of individual CG coefficients is hump-shaped in l .